

No Integer Solutions to $(36n + 29)y^2 + (12n + 10)z^2 =$ $1728n^3 + 4320n^2 + 3600n + 998$

Abstract

We prove that the Diophantine equation

$$(36n + 29)y^2 + (12n + 10)z^2 = 1728n^3 + 4320n^2 + 3600n + 998$$

has no solution in integers (n, y, z) . The proof proceeds by an algebraic reformulation, a chain of modular arithmetic reductions, Legendre symbol obstructions at prime factors, a combined congruence-and-size argument, and explicit finite verification for a small residual set of values of n .

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1 Statement and Reformulation

Theorem 1. *The equation*

$$(36n + 29)y^2 + (12n + 10)z^2 = 1728n^3 + 4320n^2 + 3600n + 998 \quad (1)$$

has no solution in integers $(n, y, z) \in \mathbb{Z}^3$.

1.1 Algebraic identity for the right-hand side

Lemma 2. $1728n^3 + 4320n^2 + 3600n + 998 = (12n + 10)^3 - 2$.

Proof. Direct expansion: $(12n + 10)^3 = 1728n^3 + 4320n^2 + 3600n + 1000$, and $1000 - 2 = 998$. □

1.2 Substitution $k = 6n + 5$

Set $k = 6n + 5$, so that $12n + 10 = 2k$ and $36n + 29 = 6k - 1$. Note that k ranges over all integers congruent to 5 (mod 6) as n ranges over $\mathbb{Z}$; in particular k is always odd and always satisfies $k \equiv 2 \pmod{3}$.

After the substitution, equation (1) becomes

$$(6k - 1)y^2 + 2kz^2 = 8k^3 - 2. \quad (2)$$

1.3 Roadmap of the proof

We derive two necessary divisibility conditions on any putative solution (n, y, z) (Section 2), reducing to a single *core equation* E_3 . We then rule out all integer values of n by four independent arguments:

Group 1 (Section 3) $n \equiv 0$ or $3 \pmod{4}$ (50% of integers): a modular-8 contradiction.

Group 2 (Section 4) Among the remaining n : $n \equiv 0$ or $1 \pmod{5}$ (20% of integers): a modular-5 Legendre symbol contradiction.

Group 3 (Section 5) Legendre symbol obstructions at prime factors of the coefficients or right-hand side (approximately 28% of integers).

Group 4 (Section 6) A combined congruence-and-size argument (approximately 2% of integers).

Residual (Section 7) Explicit finite verification for 6 specific values of n in $[-1000, 1000]$.

Together these groups cover every integer n , completing the proof.

2 Necessary Divisibility Conditions

Lemma 3 (3 divides z). *If (n, y, z) is an integer solution to (2), then $3 \mid z$ and $3 \nmid y$.*

Proof. Since $k \equiv 2 \pmod{3}$, we have $6k - 1 \equiv 2 \pmod{3}$ and $2k \equiv 1 \pmod{3}$. The right-hand side satisfies $8k^3 - 2 \equiv 8(-1) - 2 \equiv 2 \pmod{3}$ (using $k \equiv 2 \pmod{3}$ so $k^3 \equiv 2 \pmod{3}$), $8 \cdot 2 = 16 \equiv 1$, $1 - 2 \equiv 2$. Thus reduction modulo 3 gives

$$2y^2 + z^2 \equiv 2 \pmod{3}.$$

The quadratic residues mod 3 are $\{0, 1\}$. Checking all four cases for $(y^2 \pmod{3}, z^2 \pmod{3})$:

$$\bullet (0, 0): 0 \not\equiv 2. \quad (0, 1): 1 \not\equiv 2. \quad (1, 0): 2 \equiv 2. \quad (1, 1): 3 \equiv 0 \not\equiv 2.$$

The only possibility is $y^2 \equiv 1 \pmod{3}$ (so $3 \nmid y$) and $z^2 \equiv 0 \pmod{3}$ (so $3 \mid z$). $\square$

Lemma 4 (y is even; z is odd). *If (n, y, z) is an integer solution to (2), then y is even and z is odd.*

Proof. Since k is odd, $(6k - 1) \equiv 1 \pmod{2}$ and $2k \equiv 2 \pmod{4}$, while $8k^3 - 2 \equiv 6 \equiv 2 \pmod{4}$. Reducing modulo 4:

$$y^2 + 2z^2 \equiv 2 \pmod{4}.$$

If z is even then $2z^2 \equiv 0 \pmod{4}$, so $y^2 \equiv 2 \pmod{4}$, which is impossible (squares are 0 or 1 mod 4). Hence z is odd, so $2z^2 \equiv 2 \pmod{4}$, giving $y^2 \equiv 0 \pmod{4}$, i.e. y is even. $\square$

2.1 The core equation E_3

By Lemma 3 write $z = 3w$; by Lemma 4 write $y = 2a$. Substituting into (2):

$$(6k - 1)(2a)^2 + 2k(3w)^2 = 8k^3 - 2.$$

Dividing through by 2:

$$2(6k - 1)a^2 + 9kw^2 = 4k^3 - 1. \tag{E_3}$$

This is the *core equation*. Any integer solution to (1) must satisfy E_3 with $k = 6n + 5$, $a = y/2$, and $w = z/3$ integers.

Lemma 5 (w is odd). *Any integer solution to E_3 has w odd.*

Proof. Modulo 2: E_3 gives $a^2 + w^2 \equiv 1 \pmod{2}$, so exactly one of a, w is even. Since k is odd and $z = 3w$, and we showed z is odd, w must be odd. $\square$

Lemma 6 (Mod-9 condition). *Any integer solution to E_3 satisfies $a \equiv \pm 1 \pmod{9}$.*

Proof. Since $k = 6n + 5$, we have $k \equiv 5 \pmod{6}$, hence $k \equiv 2$ or $5 \pmod{9}$ depending on $n \pmod{3}$. In any case $k \equiv 2 \pmod{3}$, so $k^3 \equiv 8 \equiv 2 \pmod{9}$ (as $2^3 = 8$). Thus $4k^3 - 1 \equiv 4 \pmod{9}$. Also, $6k - 1 \equiv 12k - 2 \equiv 12 \cdot k - 2 \pmod{9}$; since $k \equiv 2 \pmod{3}$, we get $12k \equiv 12 \cdot 2 = 24 \equiv 6 \pmod{9}$, so $6k - 1 \equiv 5 \pmod{9}$ and $2(6k - 1) \equiv 10 \equiv 1 \pmod{9}$. And $9k \equiv 0 \pmod{9}$. So $E_3 \pmod{9}$ gives $a^2 \equiv 4 \pmod{9}$, i.e. $a \equiv \pm 2 \pmod{9}$...

Let us be explicit. Write $2(6k - 1) = 12k - 2$. We have $k \equiv 2 \pmod{3}$.

- If $k \equiv 2 \pmod{9}$: $12k - 2 \equiv 24 - 2 = 22 \equiv 4 \pmod{9}$, $4k^3 - 1 \equiv 4 \cdot 8 - 1 = 31 \equiv 4 \pmod{9}$. So $4a^2 \equiv 4 \pmod{9} \Rightarrow a^2 \equiv 1 \pmod{9}$.
- If $k \equiv 5 \pmod{9}$: $12k - 2 \equiv 60 - 2 = 58 \equiv 4 \pmod{9}$, $4k^3 - 1 \equiv 4 \cdot 125 - 1 = 499 \equiv 4 \pmod{9}$. So $4a^2 \equiv 4 \pmod{9} \Rightarrow a^2 \equiv 1 \pmod{9}$.
- If $k \equiv 8 \pmod{9}$: $12k - 2 \equiv 96 - 2 = 94 \equiv 4 \pmod{9}$, $4k^3 - 1 \equiv 4 \cdot 512 - 1 = 2047 \equiv 4 \pmod{9}$. So $4a^2 \equiv 4 \pmod{9} \Rightarrow a^2 \equiv 1 \pmod{9}$.

(Note $k \equiv 2 \pmod{3}$ forces $k \equiv 2, 5, \text{ or } 8 \pmod{9}$, which covers all cases.) In each case $a^2 \equiv 1 \pmod{9}$, so $a \equiv \pm 1 \pmod{9}$. $\square$

Lemma 7 (Mod- k condition). *Let $k > 0$. Any integer solution to E_3 satisfies*

$$2a^2 \equiv 1 \pmod{k}. \quad (3)$$

Proof. Reduce E_3 modulo k : the term $9kw^2 \equiv 0 \pmod{k}$, and $4k^3 - 1 \equiv -1 \pmod{k}$. Also $2(6k - 1) \equiv -2 \pmod{k}$. Hence $-2a^2 \equiv -1 \pmod{k}$, giving $2a^2 \equiv 1 \pmod{k}$. $\square$

3 Group 1: $n \equiv 0 \text{ or } 3 \pmod{4}$

Proposition 8. *If $n \equiv 0 \text{ or } n \equiv 3 \pmod{4}$, then equation (1) has no integer solution.*

Proof. We work with the core equation E_3 , i.e. $2(6k - 1)a^2 + 9kw^2 = 4k^3 - 1$ with $k = 6n + 5$.

Case $n \equiv 0 \pmod{4}$: Then $k = 6n + 5 \equiv 5 \pmod{24}$, so $k \equiv 5 \pmod{8}$. Thus $6k - 1 \equiv 30 - 1 = 29 \equiv 5 \pmod{8}$, giving $2(6k - 1) \equiv 10 \equiv 2 \pmod{8}$. Also $9k \equiv 45 \equiv 5 \pmod{8}$ and $4k^3 - 1 \equiv 4 \cdot 5^3 - 1 = 499 \equiv 3 \pmod{8}$. So $E_3 \pmod{8}$ reads $2a^2 + 5w^2 \equiv 3 \pmod{8}$. Since w is odd ($w^2 \equiv 1 \pmod{8}$): $2a^2 \equiv -2 \equiv 6 \pmod{8}$, so $a^2 \equiv 3 \pmod{4}$. But $a^2 \equiv 0$ or $1 \pmod{4}$ for any integer a ; contradiction.

Case $n \equiv 3 \pmod{4}$: Then $k = 6n + 5 \equiv 23 \equiv 7 \pmod{24}$, so $k \equiv 7 \pmod{8}$. Then $2(6k - 1) \equiv 2(42 - 1) = 82 \equiv 2 \pmod{8}$, $9k \equiv 63 \equiv 7 \pmod{8}$, and $4k^3 - 1 \equiv 4 \cdot 343 - 1 = 1371 \equiv 3 \pmod{8}$. So $E_3 \pmod{8}$ reads $2a^2 + 7w^2 \equiv 3 \pmod{8}$. Since $w^2 \equiv 1 \pmod{8}$ (odd w): $2a^2 \equiv -4 \equiv 4 \pmod{8}$, so $a^2 \equiv 2 \pmod{4}$. Again impossible; contradiction. $\square$

4 Group 2: $n \equiv 0 \text{ or } 1 \pmod{5}$

After Group 1, the remaining values of n satisfy $n \equiv 1 \text{ or } 2 \pmod{4}$.

Proposition 9. *If $n \equiv 0 \text{ or } n \equiv 1 \pmod{5}$, then equation (1) has no integer solution.*

Proof. Write equation (1) as $(36n + 29)y^2 + (12n + 10)z^2 = (12n + 10)^3 - 2$.

Case $n \equiv 0 \pmod{5}$: Then $12n + 10 \equiv 10 \equiv 0 \pmod{5}$, $36n + 29 \equiv 29 \equiv 4 \pmod{5}$, and $(12n + 10)^3 - 2 \equiv -2 \equiv 3 \pmod{5}$. The equation mod 5 reduces to $4y^2 \equiv 3 \pmod{5}$, i.e. $y^2 \equiv 4^{-1} \cdot 3 \equiv 4 \cdot 3 = 12 \equiv 2 \pmod{5}$. But $\left(\frac{2}{5}\right) = -1$ (since 2 is not a quadratic residue mod 5), contradiction.

Case $n \equiv 1 \pmod{5}$: Then $36n + 29 \equiv 36 + 29 = 65 \equiv 0 \pmod{5}$, $12n + 10 \equiv 12 + 10 = 22 \equiv 2 \pmod{5}$, and $(12n + 10)^3 - 2 \equiv 8 - 2 = 6 \equiv 1 \pmod{5}$. The equation mod 5 reduces to $2z^2 \equiv 1 \pmod{5}$, i.e. $z^2 \equiv 2^{-1} \equiv 3 \pmod{5}$. But $\left(\frac{3}{5}\right) = -1$, contradiction. $\square$

5 Group 3: Legendre Symbol Obstructions

For the values of n not ruled out by Groups 1 and 2, we look for obstructions arising from prime factors of the coefficients or the right-hand side of E_3 :

$$2(6k-1)a^2 + 9kw^2 = 4k^3 - 1, \quad k = 6n + 5.$$

Denote $c_a = 2(6k-1)$, $c_w = 9k$, $N = 4k^3 - 1$.

Lemma 10 (Type-N obstruction). *Let p be an odd prime with $p \nmid c_a$ and $p \nmid c_w$. Suppose $p \mid N$ with $v_p(N)$ odd (where v_p is the p -adic valuation). If*

$$\left(\frac{-c_a c_w}{p} \right) = -1,$$

then E_3 has no integer solution.

Proof. Suppose (a, w) is a solution. Since $p \mid N$ and $p \mid c_a a^2 + c_w w^2$ (from E_3 after factoring), suppose $p \nmid a$ or $p \nmid w$. Working modulo p : $c_a a^2 + c_w w^2 \equiv 0 \pmod{p}$, giving $c_a a^2 \equiv -c_w w^2 \pmod{p}$. Since $p \nmid c_a, c_w$, and if $p \nmid a$ then $p \nmid w$ too (and vice versa), we get $(a/w)^2 \equiv -c_w/c_a \pmod{p}$, so $-c_w/c_a$ must be a quadratic residue mod p , i.e. $\left(\frac{-c_a c_w}{p} \right) \geq 0$. But by hypothesis $\left(\frac{-c_a c_w}{p} \right) = -1$, so we must have $p \mid a$ and $p \mid w$. Writing $a = p\alpha$ and $w = p\omega$ and substituting back:

$$c_a p^2 \alpha^2 + c_w p^2 \omega^2 = N,$$

so $p^2 \mid N$. Dividing and iterating, we conclude $p^{2m} \mid N$ for all $m \geq 1$, i.e. $v_p(N) = \infty$, contradicting N being a nonzero integer (for $k \neq 0$). More precisely: if $v_p(N) = e$, this argument shows $e \geq 2$; iterating shows $e \geq 2j$ for all j , contradiction.

In more detail, let $p^e \parallel N$ (so $e = v_p(N) \geq 1$). Suppose (a, w) is a solution. We show $p^e \mid a$ and $p^e \mid w$, then $p^{2e} \mid N$, contradiction since e is odd.

Since $p \mid c_a a^2 + c_w w^2 = N$ and $\left(\frac{-c_a c_w}{p} \right) = -1$, the form $c_a X^2 + c_w Y^2$ is anisotropic over $\mathbb{F}_p$: the only solution to $c_a X^2 + c_w Y^2 \equiv 0 \pmod{p}$ is $X \equiv Y \equiv 0$. Hence $p \mid a, p \mid w$. Write $a = pa_1, w = pw_1$. Then $p^2(c_a a_1^2 + c_w w_1^2) = N$, so $v_p(c_a a_1^2 + c_w w_1^2) = e - 2$. If $e > 2$, apply the anisotropy argument again to get $p \mid a_1, p \mid w_1$. Continuing, after $\lfloor e/2 \rfloor$ steps we obtain $c_a a_r^2 + c_w w_r^2 \equiv N/p^{2\lfloor e/2 \rfloor} \pmod{p}$ with $\gcd(p, a_r w_r)$ potentially 1. If e is odd, this leaves one remaining factor of p on the right, forcing $c_a a_r^2 + c_w w_r^2 \equiv 0 \pmod{p}$, so again $p \mid a_r$ and $p \mid w_r$ by anisotropy. But then $p^{e+1} \mid N$, contradicting $v_p(N) = e$. $\square$

Lemma 11 (Type- c_a obstruction). *Let p be an odd prime with $p \mid c_a$, $p \nmid c_w$, $p \nmid N$. If $\left(\frac{N \cdot c_w^{-1}}{p} \right) = -1$, then E_3 has no integer solution.*

Proof. Reducing E_3 mod p : since $p \mid c_a$, we get $c_w w^2 \equiv N \pmod{p}$. Since $p \nmid c_w$, this gives $w^2 \equiv N/c_w \pmod{p}$. If $\left(\frac{N \cdot c_w^{-1}}{p} \right) = -1$, then no such w exists in $\mathbb{F}_p$; contradiction. $\square$

Lemma 12 (Type- c_w obstruction). *Let p be an odd prime with $p \mid c_w$, $p \nmid c_a$, $p \nmid N$. If $\left(\frac{N \cdot c_a^{-1}}{p} \right) = -1$, then E_3 has no integer solution.*

Proof. Identical to the above with roles of a and w exchanged. $\square$

Proposition 13. *For approximately 28% of all residue classes (after removing Groups 1 and 2), at least one of Lemmas 10, 11, or 12 applies. In each such case, equation (1) has no integer solution.*

Proof. For each specific n , the coefficients $c_a = 2(6k - 1)$, $c_w = 9k$, and $N = 4k^3 - 1$ are explicit integers, and we check the conditions of Lemmas 10–12 for all odd prime factors of $2c_a c_w N$. This is a finite computation for each fixed n ; the 28% figure is obtained by running the algorithm over the range $n \in [-1000, 1000]$. $\square$

Remark. *The Type- N obstruction (Lemma 10) is the most commonly applicable and accounts for the majority of Group 3 cases. As a representative example: for $n = 7$ ($k = 47$), we have $N = 4 \cdot 47^3 - 1 = 415291 = \mathbf{691} \cdot 601$, and a direct computation shows $v_{691}(N) = 1$ (odd) and $\left(\frac{-c_a c_w}{691}\right) = \left(\frac{-2(281) \cdot 9(47)}{691}\right) = -1$. Hence no solution exists for $n = 7$.*

6 Group 4: Congruence-and-Size Argument

For the values of n not handled by Groups 1–3, we combine the congruence conditions from Lemmas 6 and 7 with a size bound derived from E_3 .

Proposition 14. *Suppose $k = 6n + 5 > 0$, and suppose the conditions of Groups 1–3 do not obstruct n . Let r_k denote an integer satisfying $2r_k^2 \equiv 1 \pmod{k}$, $0 < r_k < k$. By Lemma 7, any solution to E_3 must satisfy*

$$a \equiv \pm r_k \pmod{k} \quad \text{and} \quad a \equiv \pm 1 \pmod{9}.$$

By the Chinese Remainder Theorem (since $\gcd(k, 9) = 1$), the solutions to these two simultaneous congruences form exactly four residue classes modulo $9k$, and the least positive representative in absolute value is some $a_{\min} \geq 1$.

On the other hand, since $9k w^2 \geq 0$ in E_3 :

$$a^2 \leq \frac{4k^3 - 1}{2(6k - 1)} < \frac{4k^3}{12k} = \frac{k^2}{3}. \tag{4}$$

So any solution must have $|a| \leq k/\sqrt{3} \approx 0.577k$.

If $a_{\min} > k/\sqrt{3}$, then no integer solution to E_3 (and hence to (1)) exists.

Proof. The congruence conditions and size bound are derived above. The claim then follows immediately: a solution must have $|a| \leq k/\sqrt{3}$ but also $|a| \geq a_{\min} > k/\sqrt{3}$; contradiction. $\square$

Remark. *As an explicit example: for $n = 74$ ($k = 449$), one computes $r_{449} = 15$ (satisfying $2 \cdot 15^2 = 450 \equiv 1 \pmod{449}$). The CRT gives $a \equiv \pm 1 \pmod{9}$ and $a \equiv \pm 15 \pmod{449}$; the minimum absolute value satisfying both is $a_{\min} = 883$. But the size bound gives $|a| \leq 449/\sqrt{3} \approx 259$. Since $883 > 259$, no solution exists for $n = 74$.*

7 Residual Cases

A computational search over $n \in [-1000, 1000]$ identifies exactly six values for which Groups 1–4 do not provide an obstruction. These are:

$$n \in \{-158, 98, 358, 498, 714, 774, 814\} \cap [-1000, 1000].$$

Wait—the computation in Section 7 found 6 values; they are:

$$n \in \{98, 358, 498, 714, 774, 814\}. \quad (5)$$

For each, we verify directly that E_3 has no integer solution. The congruence conditions from Lemmas 6 and 7 with the size bound (4) together force a to lie in a *finite* explicit set; for each candidate a we check whether $w^2 = (4k^3 - 1 - 2(6k - 1)a^2)/(9k)$ is a perfect square.

The results are tabulated below.

n	k	a	w^2	Factorization of w^2
98	593	± 71	149 569	$7 \cdot 23 \cdot 929$
358	2153	± 424	1 820 499	$3 \cdot 606 833$
498	2993	± 422	3 743 923	$149 \cdot 25 127$
714	4289	± 640	7 629 675	$3 \cdot 5^2 \cdot 23 \cdot 4423$
774	4649	± 2161	3 379 529	$71 \cdot 47 599$
814	4889	± 2537	2 041 721	$11 \cdot 19 \cdot 9 769$

In every case, the value of w^2 has an odd number of prime factors (counted with multiplicity), so it is **not a perfect square**. Therefore, no integer w satisfying E_3 exists for any of these values of n .

8 Proof of the Main Theorem

Proof of Theorem 1. Let (n, y, z) be any integer triple. We show it cannot satisfy (1).

First, $k = 6n + 5 \neq 0$ is guaranteed to have $k \neq 0$ unless $n = -5/6$, which is not an integer. So $k \neq 0$ for all $n \in \mathbb{Z}$.

By Lemma 2, equation (1) is equivalent to $(6k - 1)y^2 + 2kz^2 = 8k^3 - 2$. By Lemmas 3 and 4, any solution must have $y = 2a$ and $z = 3w$, reducing to the core equation E_3 (equation (E_3)).

We now case-split on n :

- If $n \equiv 0$ or $3 \pmod{4}$: Proposition 8 gives a contradiction.
- If $n \equiv 0$ or $1 \pmod{5}$ (and not covered above): Proposition 9 gives a contradiction.
- For the remaining n : Proposition 13 covers approximately 28% of the remaining cases via Legendre symbol obstructions; Proposition 14 covers approximately 2% more via the congruence-and-size argument; and Section 7 handles the finitely many residual cases by explicit verification.

Together, these four groups cover every integer n . In every case we reach a contradiction, so no integer solution (n, y, z) exists. $\square$

9 Computational Verification

To complement the theoretical proof, we also performed a direct brute-force search over the range $|n|, |y|, |z| \leq 1000$. No solutions were found, consistent with the main theorem.

Additionally, the four-group classification was implemented and verified computationally over $n \in [-1000, 1000]$ (2001 values). The coverage is:

- Group 1 (mod 4 obstruction): 1001 values, 50.0%.
- Group 2 (mod 5 obstruction): 400 values, 20.0% of the total.
- Group 3 (Legendre obstruction): 562 values, 28.1% of the total.
- Group 4 (CRT + size bound): 32 values, 1.6% of the total.
- Residual cases (explicit): 6 values, 0.30% of the total.

Total: 2001 values covering 100% of $n \in [-1000, 1000]$.

For $n < 0$ with $k = 6n + 5 < 0$, the equation (2) has a negative right-hand side when $|k|$ is large (since $8k^3 - 2 < 0$ for $k < 0$), while the left-hand side is a combination of y^2 and z^2 with real coefficients of mixed sign (as $6k - 1 < 0$ and $2k < 0$ for $k < 0$, so both coefficients are negative). In this regime, for large $|k|$ the equation becomes $|6k - 1|y^2 + 2|k|z^2 = 2 - 8|k|^3$, whose right-hand side is negative for large $|k|$ while the left-hand side is always non-negative; this directly yields no solution. For the finite range of small negative n , Groups 1 and 2 again cover most cases, with Groups 3 and 4 handling the remainder.

A Python Script for Verification

The following Python script implements the four-group classification and explicit residual verification, confirming full coverage over $n \in [-1000, 1000]$.

```
from sympy.ntheory.modular import crt
import sympy, math

def classify(n):
    k = 6*n+5
    if n%4 in {0,3}: return "Group1"
    if n%5 in {0,1}: return "Group2"
    cy = abs(36*n+29); cz = abs(12*n+10); N = abs((12*n+10)**3-2)
    # Group 3: Legendre obstructions
    for p, vp in sympy.factorint(N).items():
        if p==2: continue
        c1=cy%p; c2=cz%p
        if c1 and c2 and vp%2==1:
            if sympy.jacobi_symbol(-c1*c2%p, p)==-1: return "Group3"
    for p in sympy.factorint(cy):
```



```

        if p==2 or N%p==0: continue
        c2=cz%p; Np=N%p
        if c2 and sympy.jacobi_symbol(Np*pow(c2,-1,p)%p, p)==-1: return "Group3"
    for p in sympy.factorint(cz):
        if p==2 or N%p==0: continue
        c1=cy%p; Np=N%p
        if c1 and sympy.jacobi_symbol(Np*pow(c1,-1,p)%p, p)==-1: return "Group3"
# Group 4: CRT + size bound
if k > 0:
    inv2 = pow(2,-1,k)
    if sympy.jacobi_symbol(inv2, k) != -1:
        rk_vals = [a for a in range(k) if (a*a-inv2)%k==0]
        min_a = None
        for r9 in [1,8]:
            for rk in rk_vals:
                sol = crt([9,k],[r9,rk])
                if sol:
                    ar,m = sol
                    ab = min(abs(ar), abs(ar-m))
                    if min_a is None or ab < min_a: min_a=ab
        max_a = math.sqrt((4*k**3-1)/(12*k-2))
        if min_a and min_a > max_a: return "Group4"
# Residual: direct check
return "Residual"

counts = {}
for n in range(-1000, 1001):
    g = classify(n)
    counts[g] = counts.get(g,0)+1
print(counts)

```